

ECEN 454 - Lab Report

Lab Number: 2

**Lab Title: Cadence Custom Layout: Design Rules, Extraction, and
Verification**

Section Number: 341

Student's Name: Abhishek Bhattacharyya

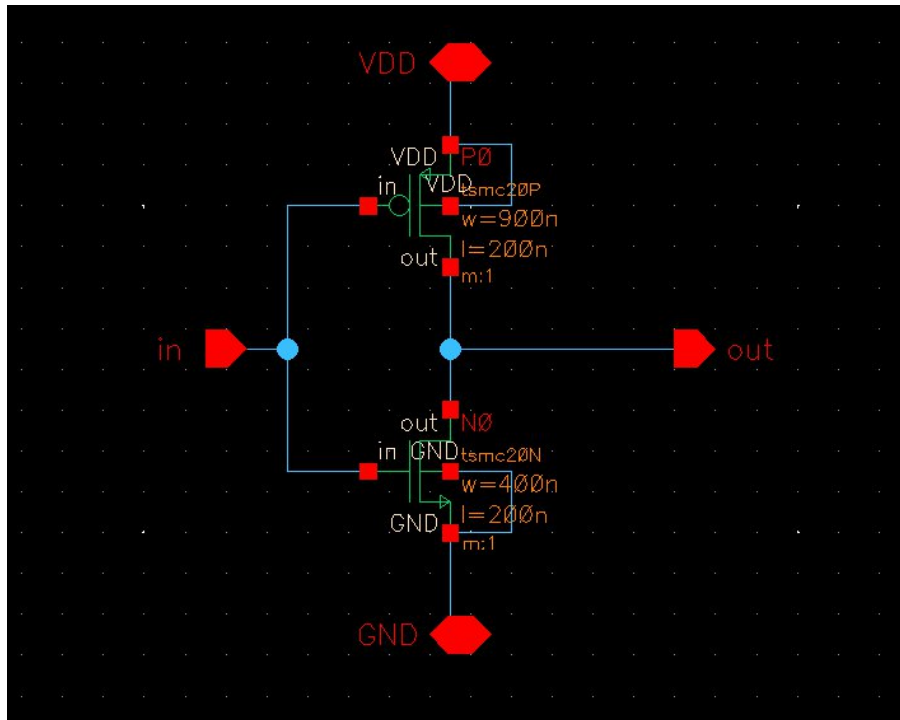
Student's UIN: 731002289

Date: 9/18/2024

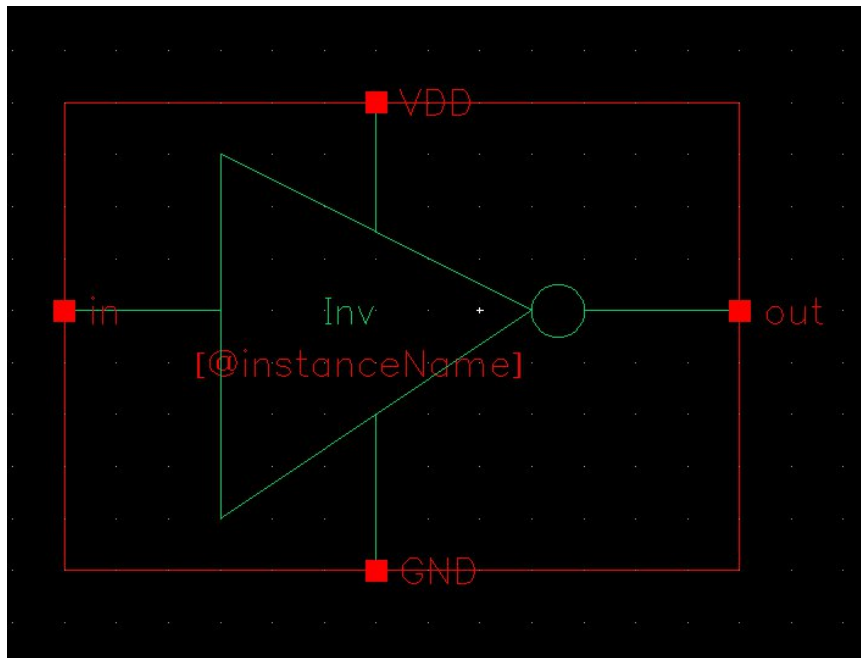
TA: Nishank Bansal

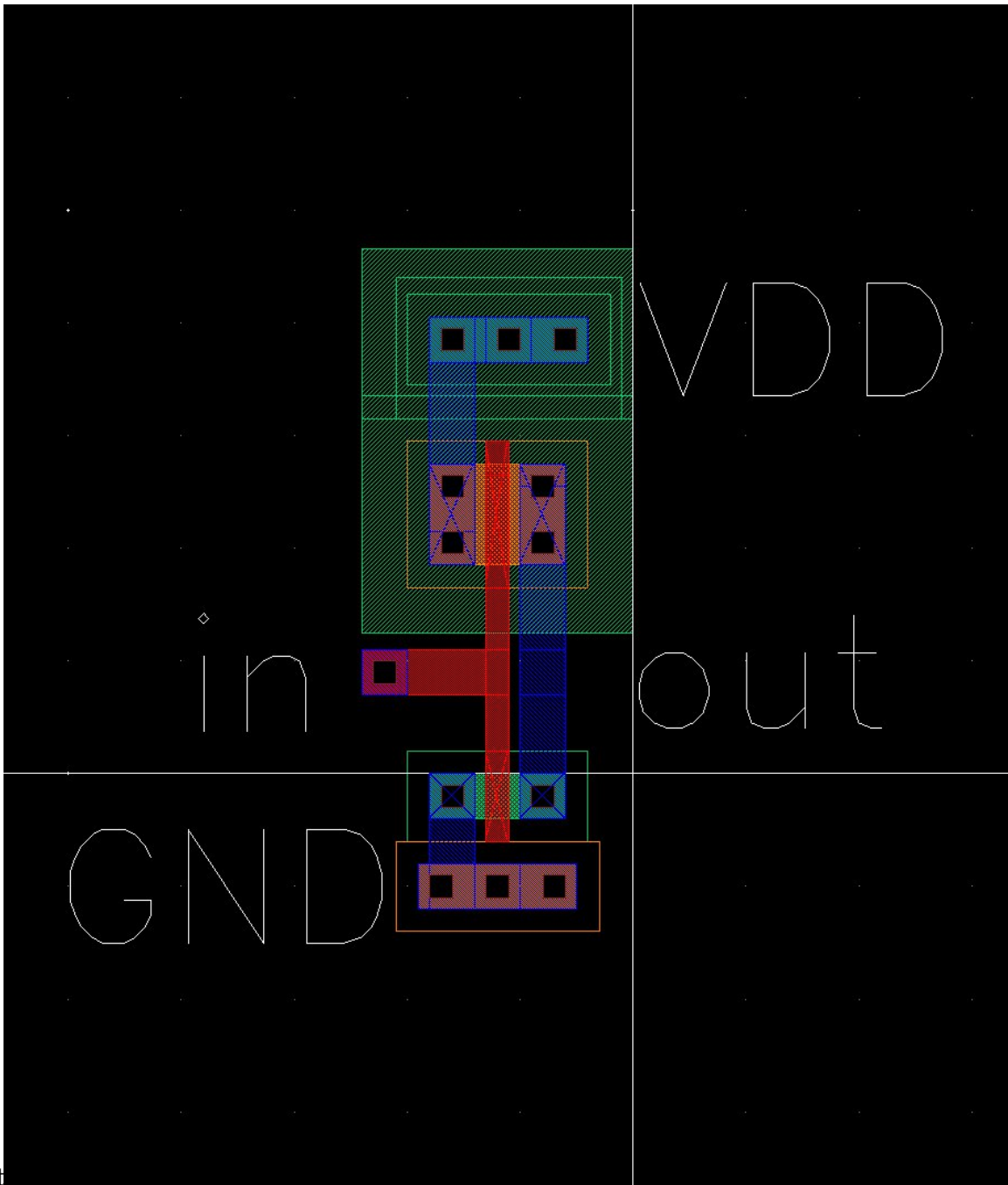
Inverter

Schematic



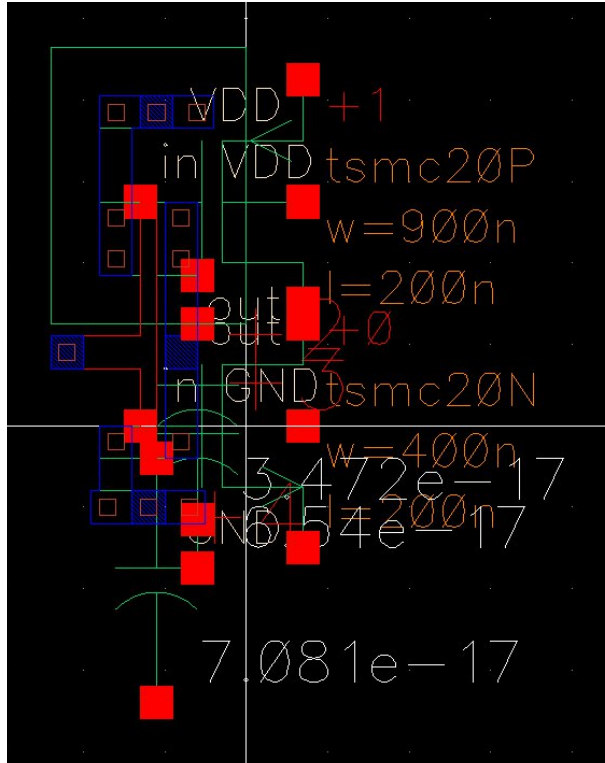
Symbol





Layout

Extracted

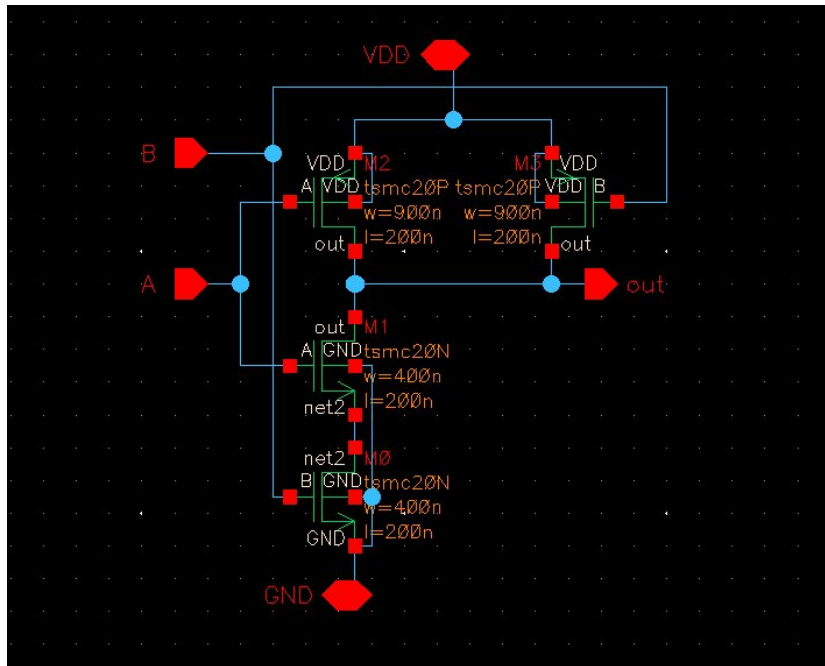


LVS Report Output

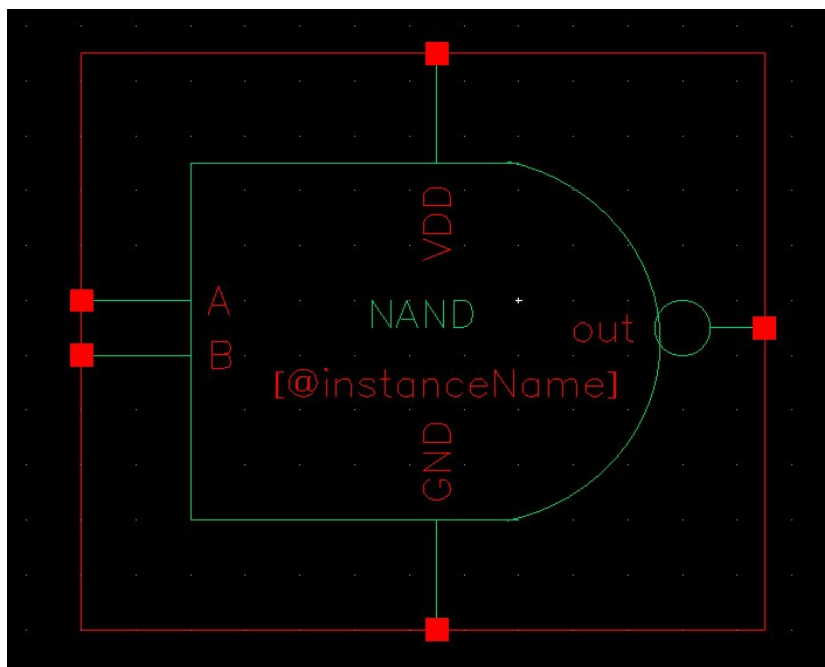
```
1.5 svveter_jvs.txt 1.5 NAND_LVS.txt 1.5 NOR_LVS.txt
1 @(#) $CDS: LVS version 6.1.8-64b 01/30/2024 19:46 (cpgbld02) $
2
3 Command line: /opt/coe/cadence/IC618/tools.lnx86/dfII/bin/64bit/LVS -dir /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS -l -s -t
4 /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/layout /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/schematic
5 Like matching is enabled.
6 Net swapping is enabled.
7 Using terminal names as correspondence points.
8 Compiling Diva LVS rules...
9
10 Net-list summary for /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/layout/netlist
11 count
12 4 nets
13 4 terminals
14 1 pmos
15 1 nmos
16
17 Net-list summary for /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/schematic/netlist
18 count
19 4 nets
20 4 terminals
21 1 pmos
22 1 nmos
23
24 Terminal correspondence points
25 N0 N3 GND
26 N3 N2 VDD
27 N2 N0 in
28 N1 N1 out
29
30 Devices in the netlist but not in the rules:
31 pcapacitor
32 Devices in the rules but not in the netlist:
33 cap nfet pfet nmos4 pmos4
34
35 The net-lists match.
```

NAND2

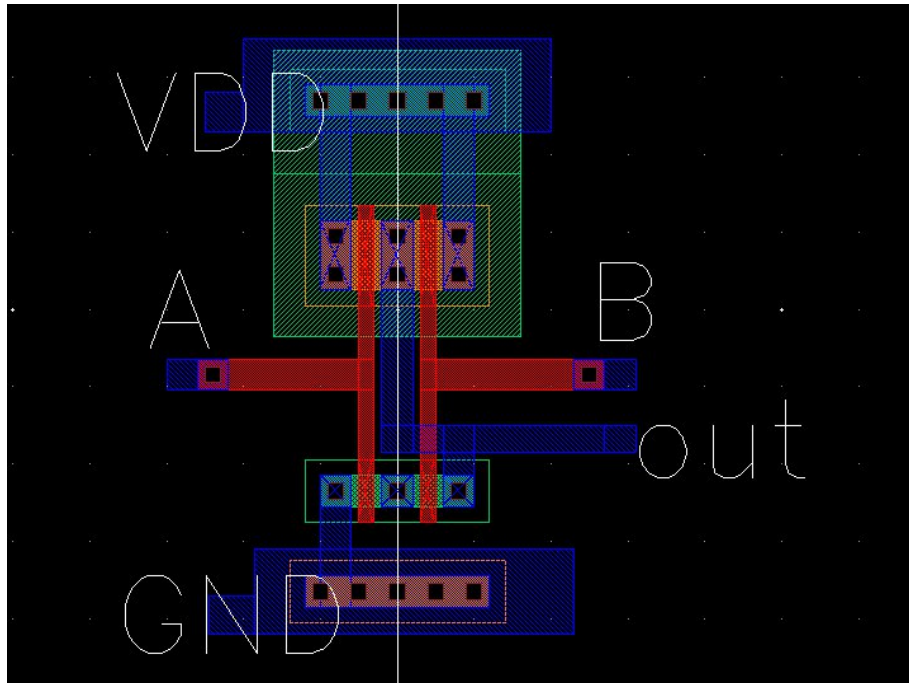
Schematic



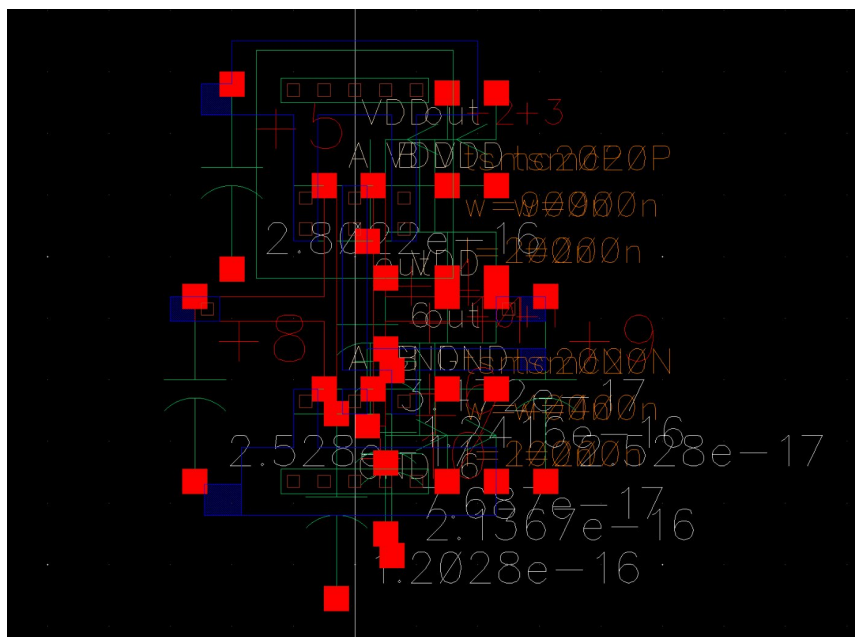
Symbol



Layout



Extracted

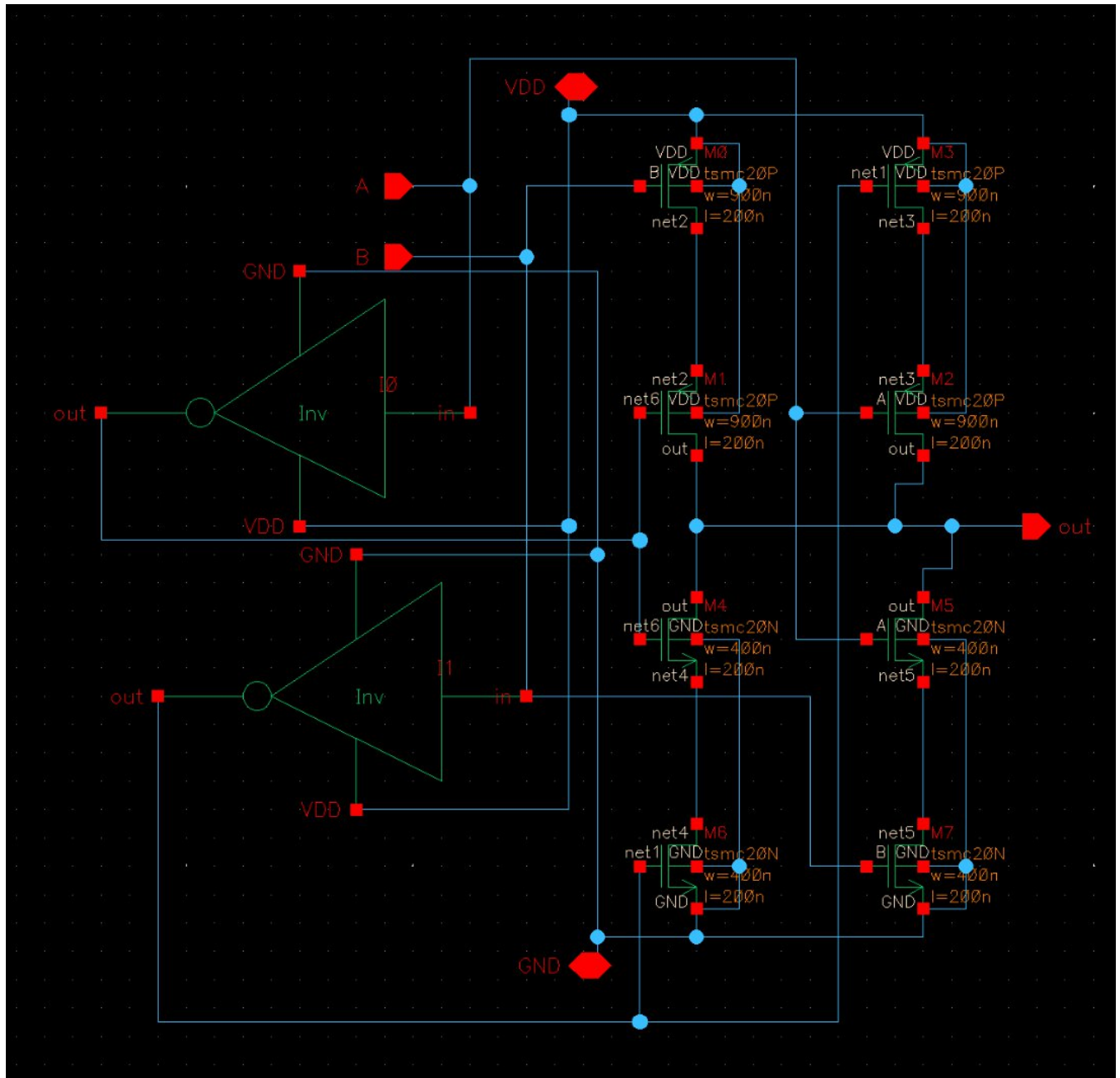


LVS Report Output

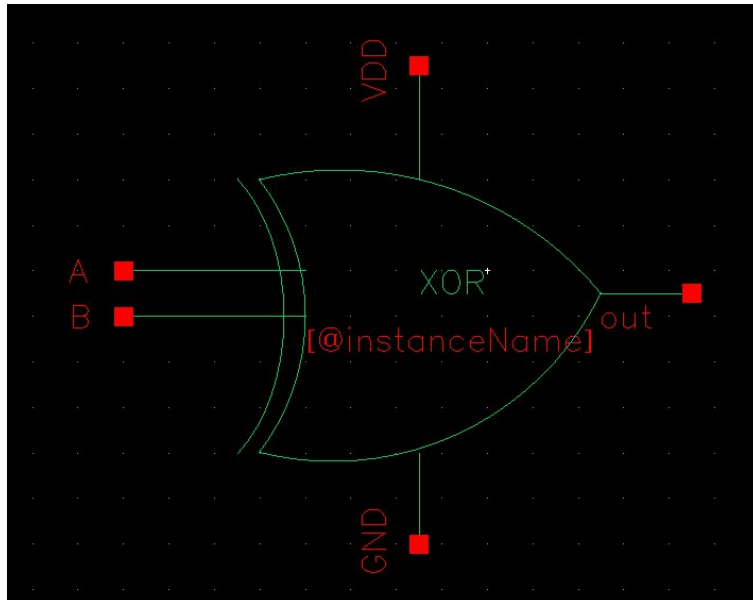
```
1 @(#) $CDS: LVS version 6.1.8-64b 01/30/2024 19:46 (cpgbld02) $
2
3 Command line: /opt/coe/cadence/IC618/tools.lnx86/dfII/bin/64bit/LVS -dir /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS -l -s -t
4 /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/layout /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/schematic
5 Like matching is enabled.
6 Net swapping is enabled.
7 Using terminal names as correspondence points.
8 Compiling Diva LVS rules...
9
10 Net-list summary for /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/layout/netlist
11   count
12   6          nets
13   5          terminals
14   2          pmos
15   2          nmos
16
17 Net-list summary for /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/schematic/netlist
18   count
19   6          nets
20   5          terminals
21   2          pmos
22   2          nmos
23
24 Terminal correspondence points
25   N3      N0      A
26   N2      N1      B
27   N1      N3      GND
28   N5      N2      VDD
29   N4      N5      out
30
31 Devices in the netlist but not in the rules:
32   pcapacitor
33 Devices in the rules but not in the netlist:
34   cap nfet pfet nmos4 pmos4
35
36 The net-lists match.
```

XOR2

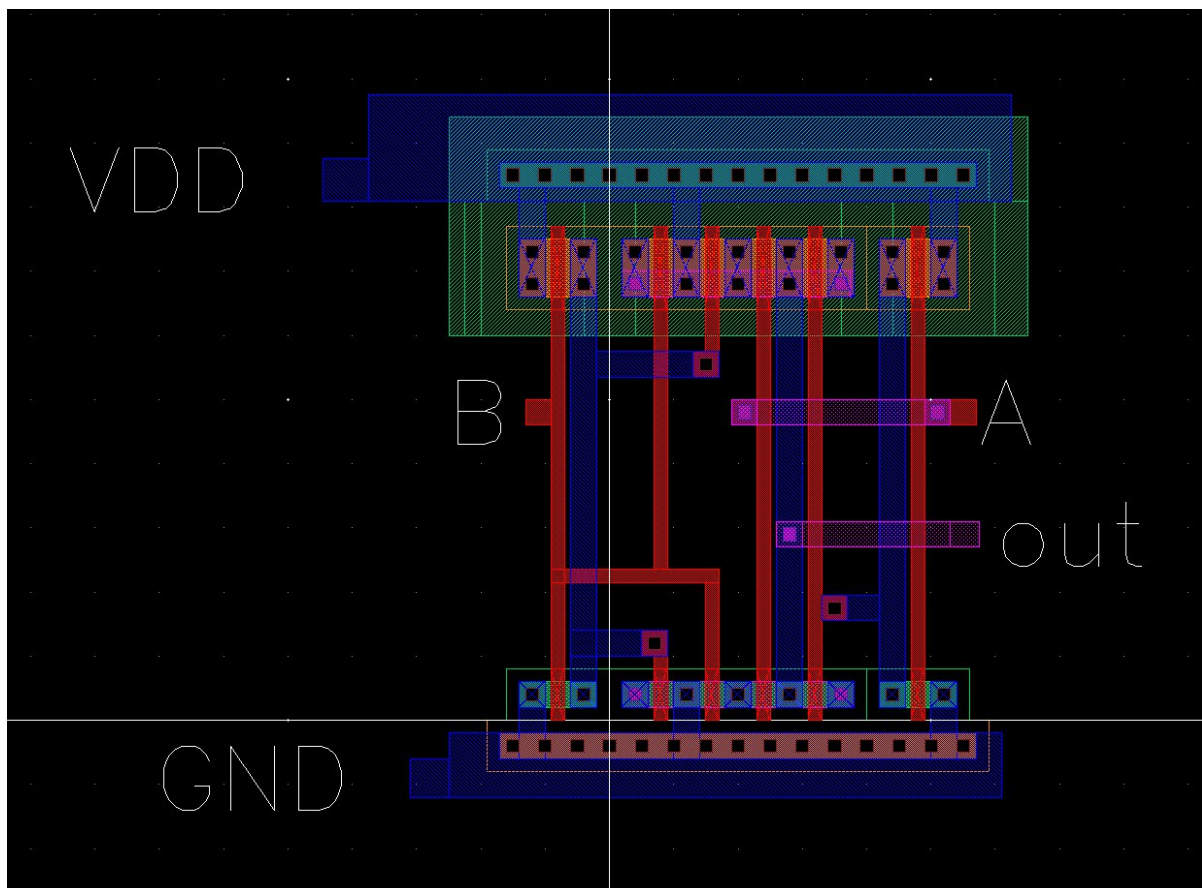
Schematic



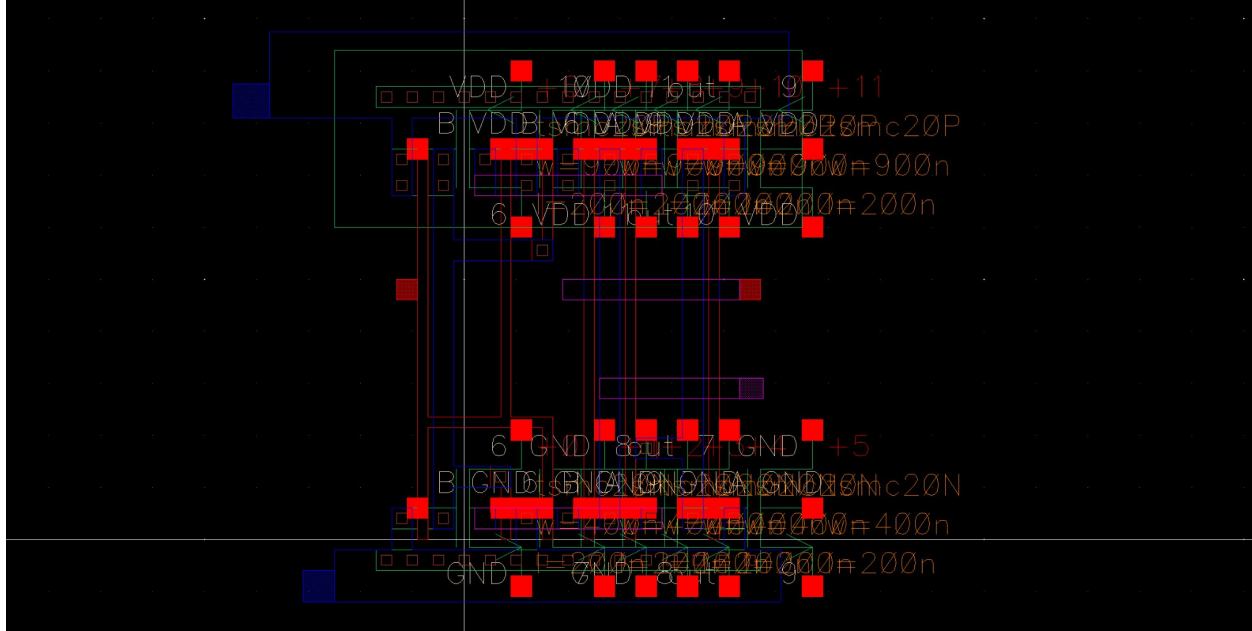
Symbol



Layout



Extracted



LVS Report Output

```
1 @ (#) $CDS: LVS version 6.1.8-64b 01/30/2024 19:46 (cpgbld02) $
2
3 Command line: /opt/coe/cadence/IC618/tools.lnx86/dfl/bin/64bit/LVS -dir /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS -l -s -t
4 /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/layout /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/schematic
5 Like matching is enabled.
6 Net swapping is enabled.
7 Using terminal names as correspondence points.
8
9 Net-list summary for /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/layout/netlist
10 count
11 11 nets
12 5 terminals
13 6 pmos
14 6 nmos
15
16 Net-list summary for /home/ugrads/a/abhi.bhattacharyya/ecen454/LVS/schematic/netlist
17 count
18 11 nets
19 5 terminals
20 6 pmos
21 6 nmos
22
23 Terminal correspondence points
24 N8 N0 A
25 N7 N1 B
26 N6 N3 GND
27 N10 N2 VDD
28 N9 N4 out
29
30 Devices in the netlist but not in the rules:
31 pmos nmos
32
33 The net-lists match.
```